DS8001 Project Proposal: Permutation-Based Approximate kNN Search for Mixed-Type Hospital Encounters

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1. Problem and questions. Exact k-nearest-neighbor (kNN) search compares a query with all N indexed encounters. Using hospital encounters only as a mixed-type retrieval benchmark (no clinical claim), we ask: at matched recall@10, does a truncated MI-File (Amato & Savino, 2008) reduce query latency and work relative to a truncated permutation scan (TPS; Chávez et al., 2008), and how does the trade-off change with N? We also measure sensitivity to pivot count n, retained rank length L and reranking cap c.

2. Methods and scope. We implement exact brute-force kNN as ground truth, TPS, and a truncated MI-File. Both approximate methods use the same random pivots; each ranks pivots by distance (ties by pivot ID), retains its top L, and assigns absent pivots rank L+1, our explicit truncation rule. Both use Spearman-footrule scores over all n pivots and rerank candidates by the true distance. TPS scans all N signatures; MI-File unions the query's L posting lists, scores each distinct candidate, and reranks up to min(c, candidates). Thus only shared-pivot objects can enter its candidate set; matching signatures isolates the effect of this access structure. This comparison builds on the course's metric-inverted-file and permutation-index topics (DS 8001, 2026). A scikit-learn BallTree (Pedregosa et al., 2011) is an exact library control on a weighted-L1 embedding of the same metric; we will check it against brute force, including ties. Build uses Nn pivot-distance evaluations and O(Nn log n) sorting; storage is O(NL) rank/posting entries. TPS scores in O(Nn); MI-File work is O(P+Un), for P postings read and U distinct candidates, plus n query-pivot and at most c reranking distances.

Distance. Six feature groups (age, utilization, admission/discharge, ICD-9 chapters, labs, medications) have equal total weight; features share their group's weight. Numeric/ordinal values use absolute differences after fixed index-fitted range scaling, without clipping; zero-range features are dropped. Nominal values use 0/1 mismatch. The selected numeric columns are complete in this dataset; categorical missing/unknown values and lab NOT_MEASURED are explicit fixed states, never pairwise-omitted. Thus the fixed weighted sum is a metric on the encoded feature states and has an exact weighted-L1 embedding for the BallTree. Out-of-index values retain their scaled differences.

3. Data and leakage control. Diabetes 130-US Hospitals (1999-2008): 101,766 encounters, 71,518 patients (Strack et al., 2014; UCI). Patient groups are split 90/5/5 into index, validation and test with a fixed seed; one query per patient is sampled. Fit scales and category vocabulary once on the full index partition and freeze them across nested N; unseen categories map to OTHER. ICD-9 chapters use a fixed published grouping. Exclude encounter/patient IDs, readmitted, race, gender and weight.

4. Experiments and decision rules. Nested index sizes N=10k, 50k and the full index; 500 validation and 500 test queries; five fixed pivot seeds; n in {32, 64}, L in {8, 16, 32}, c/N in {0.5%, 1%, 5%}. Tune on validation across the five seeds; freeze the lowest-median-latency settings meeting mean recall targets 0.80 and 0.90 before test, breaking ties by smaller index size then fewer distance calls. On identical test queries, compare TPS/MI-File and exact brute-force/BallTree controls; verify BallTree top-10 against brute force on all test queries, allowing exact-distance ties. Report latency, distinct true-distance calls (reusing query-pivot distances), TPS signatures scored, MI-File postings/candidates, index bytes, build cost and peak memory separately. Report MI-File zero-overlap misses: true top-10 neighbors excluded before reranking. Also compare recall at matched true-distance-call budgets. Run single-threaded, warmed, repeated query batches on the same machine; warm up and time five batches, report median and p95 query latency, and record hardware/software versions. Break candidate-score and exact-distance ties by encounter ID; report strict recall and recall crediting any neighbor tied at the exact rank-10 distance. Paired patient-level bootstrap intervals quantify query uncertainty; report each pivot seed separately. An unreached recall target is reported as such, not extrapolated.

5. Validation and scope. We will test patient separation, metric properties, deterministic ties, and exact agreement of the L1 embedding and BallTree with brute force on sampled cases. The contribution is a controlled empirical comparison, not a new index. One dataset and one declared distance limit generalization; conclusions concern algorithmic retrieval, not clinical similarity. Python 3.11+; code, versions, fixed seeds and experiment settings will be documented for reproduction.

References

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